NUMEK M5 · TEMPORAL RELIABILITY INTELLIGENCE

bike

Hourly observations; timezone is an explicit validation assumption, unresolved DST folds stay unresolved. Components are not declared historically available.

Completed evidence · 1 sources · 31 retained records

9 evidence-qualified finding(s) across 1 supplied source(s). Required stages completed over eligible records; behavioral search scope is disclosed separately.

Actions to investigate

Declared observation schedule has absent records

1002 expected distinct observation slots are absent; supplied null values count as occupied records. Review the declarations and bounded witnesses.

declared_violation · t5

Some declared clock values remain unresolved

Dependent checks exclude unresolved temporal cells; the report does not treat them as compliant. Review the declarations and bounded witnesses.

investigation_candidate · t1

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t7 · cnt

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t11 · casual

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t15 · registered

Interpretation boundary

- No composite score or universal readiness verdict.
- Investigation defaults are not validated defect criteria.
- Supplied evidence does not prove what a production pipeline consumed.

Generated 2026-10-10T21:16:21.095323+00:00 · Schema 1.4 · temporal_reliability_v1

Declarations

Evidence / source	Measurement or rule	Retained result
bike	Source and base task	{"format":"csv","role":"dataset","task":"dataset_profile","targets":null}
bike	observation	measurement
bike	layout	wide
bike	entity columns	[]
bike	observation keys	[]
bike	clock	{"column":"time","kind":"instant","encoding":"iso","format":null,"epoch_unit":null,"timezone":"America/New_York","timezone_column":null,"ambiguity":"unresolved","fold_column":null,"date_convention":null}
bike	schedule	{"kind":"fixed","cadence_ns":"3600000000000","anchor":"2011-01-01T00:00:00-05:00","civil_unit":null,"timezone":null,"month_end":"skip","source_id":null,"time_column":null,"entity_columns":[],"measurement_column":null,"expectation":"required_record","collection_status_column":null,"collection_active_values":["active"]}
bike	active intervals	[]
bike	cnt	{"name":"cnt","column":null,"representation":"quantitative","unit":"count","meaning":null,"past_observation":false}
bike	casual	{"name":"casual","column":null,"representation":"quantitative","unit":"count","meaning":null,"past_observation":false}
bike	registered	{"name":"registered","column":null,"representation":"quantitative","unit":"count","meaning":null,"past_observation":false}
bike	temp	{"name":"temp","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
bike	hum	{"name":"hum","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
bike	weathersit	{"name":"weathersit","column":null,"representation":"categorical","unit":null,"meaning":null,"past_observation":false}

14 of 14 metric rows shown. Presentation limits do not change analytical coverage.

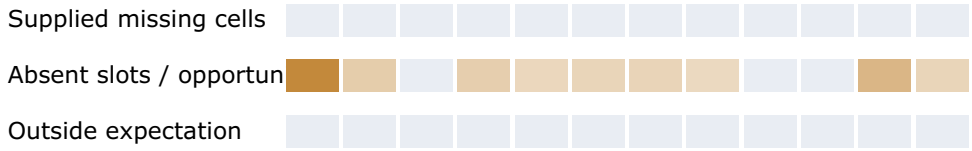
Chronology

Evidence / source	Measurement or rule	Retained result
t1 bike	rows	17379
t1 bike	cells	104274
t1 bike	declared dense missing cells	0
t1 bike	parse states / ambiguous local time	12
t1 bike	parse states / valid	104262
t1 bike	precision counts / second	104274
t1 bike	value states [casual] / measurement	casual
t1 bike	value states [casual] / state	valid
t1 bike	value states [casual] / cells	17379
t1 bike	value states [cnt] / measurement	cnt
t1 bike	value states [cnt] / state	valid
t1 bike	value states [cnt] / cells	17379
t1 bike	value states [hum] / measurement	hum
t1 bike	value states [hum] / state	valid
t1 bike	value states [hum] / cells	17379
t1 bike	value states [registered] / measurement	registered
t1 bike	value states [registered] / state	valid
t1 bike	value states [registered] / cells	17379
t1 bike	value states [temp] / measurement	temp
t1 bike	value states [temp] / state	valid

20 of 40 metric rows shown. 20 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Continuity and history

Absent observations, supplied missing values and outside-expectation periods have different meanings. Event absence alone does not prove an outage.



bike · missing values and absent slots remain separate



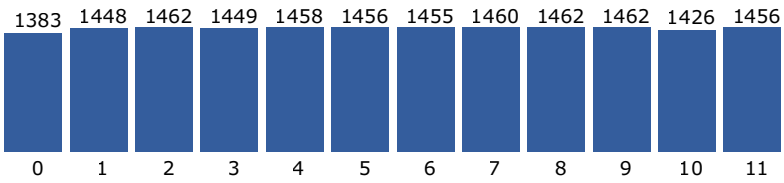
bike · expected population

Evidence / source	Measurement or rule	Retained result
t4 bike	distinct entities	1
t4 bike	distinct series	6
t4 bike	history counts / min	17377
t4 bike	history counts / max	17377
t4 bike	history counts / entity weighted mean	17377
t4 bike	history length bins [0] / observations	4
t4 bike	history length bins [0] / series	6
t4 bike	observed first tick	1293858000000000000
t4 bike	observed last tick	1357012800000000000
t4 bike	observed entry exit is lifecycle	No
t4 bike	retained histories [casual] / entity	[]
t4 bike	retained histories [casual] / measurement	casual
t4 bike	retained histories [casual] / observations	17377
t4 bike	retained histories [casual] / cells	17377
t4 bike	retained histories [casual] / first tick	1293858000000000000
t4 bike	retained histories [casual] / last tick	1357012800000000000
t4 bike	retained histories [casual] / span ticks	63154800000000000
t4 bike	retained histories [cnt] / entity	[]
t4 bike	retained histories [cnt] / measurement	cnt
t4 bike	retained histories [cnt] / observations	17377

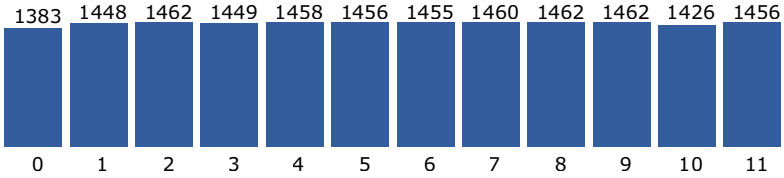
20 of 212 metric rows shown. 192 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Behavioral investigation

Thresholds and support settings are configurable investigation defaults. Reference support, sparse windows, zero dispersion, phase exclusions and population mix qualify interpretation.



bike / cnt · supplied cells per window



bike / casual · supplied cells per window

Evidence / source	Measurement or rule	Retained result
t7 bike / cnt	Reference and interpretation	{"reference_window":0,"unit":"count","meaning":"Observed effects are descriptive investigation evidence, not validated universal defects, changepoints, concept drift, or model performance."}
t7 bike / cnt	Window 0	{"window_id":0,"start_tick":"1293858000000000000","end_tick":"1299120900000000000","cells":1383,"missing":0,"invalid":0,"nonfinite":0,"finite":1383,"entities":1,"missing_rate":0.0,"quantiles":[1.0,2.0,18.0,54.0,92.0,184.89999999999986,249.5400000000002],"iqr":74.0}
t7 bike / cnt	Window 1	{"window_id":1,"start_tick":"1299120900000000000","end_tick":"1304383800000000000","cells":1448,"missing":0,"invalid":0,"nonfinite":0,"finite":1448,"entities":1,"missing_rate":0.0,"quantiles":[1.0,3.0,24.0,83.5,165.0,331.0,452.0],"iqr":141.0}
t7 bike / cnt	Window 2	{"window_id":2,"start_tick":"1304383800000000000","end_tick":"1309646700000000000","cells":1462,"missing":0,"invalid":0,"nonfinite":0,"finite":1462,"entities":1,"missing_rate":0.0,"quantiles":[2.0,5.049999999999997,57.0,170.0,294.75,479.9000000000001,579.0],"iqr":237.75}
t7 bike / cnt	Window 3	{"window_id":3,"start_tick":"1309646700000000000","end_tick":"1314909600000000000","cells":1449,"missing":0,"invalid":0,"nonfinite":0,"finite":1449,"entities":1,"missing_rate":0.0,"quantiles":[4.0,6.0,60.0,163.0,280.0,473.7999999999997,580.0799999999999],"iqr":220.0}
t7 bike / cnt	Window 4	{"window_id":4,"start_tick":"1314909600000000000","end_tick":"1320172500000000000","cells":1458,"missing":0,"invalid":0,"nonfinite":0,"finite":1458,"entities":1,"missing_rate":0.0,"quantiles":[2.0,5.0,43.0,143.5,253.0,456.0,584.4300000000001],"iqr":210.0}
t7 bike / cnt	Window 5	{"window_id":5,"start_tick":"1320172500000000000","end_tick":"1325435400000000000","cells":1456,"missing":0,"invalid":0,"nonfinite":0,"finite":1456,"entities":1,"missing_rate":0.0,"quantiles":[2.0,4.0,27.75,105.0,195.0,366.75,467.45000000000005],"iqr":167.25}
t7 bike / cnt	Window 6	{"window_id":6,"start_tick":"1325435400000000000","end_tick":"1330698300000000000","cells":1455,"missing":0,"invalid":0,"nonfinite":0,"finite":1455,"entities":1,"missing_rate":0.0,"quantiles":[1.0,3.0,33.0,111.0,207.5,422.29999999999995,527.46],"iqr":174.5}
t7 bike / cnt	Window 7	{"window_id":7,"start_tick":"1330698300000000000","end_tick":"1335961200000000000","cells":1460,"missing":0,"invalid":0,"nonfinite":0,"finite":1460,"entities":1,"missing_rate":0.0,"quantiles":[2.0,4.0,52.75,200.5,349.5,654.0,757.8200000000002],"iqr":296.75}
t7 bike / cnt	Window 8	{"window_id":8,"start_tick":"1335961200000000000","end_tick":"1341224100000000000","cells":1462,"missing":0,"invalid":0,"nonfinite":0,"finite":1462,"entities":1,"missing_rate":0.0,"quantiles":[4.0,8.0,73.25,243.0,415.0,680.95,825.1700000000003],"iqr":341.75}

Evidence / source	Measurement or rule	Retained result
t7 bike / cnt	Window 9	{"window_id":9,"start_tick":"1341224100000000000","end_tick":"1346487000000000000","cells":1462,"missing":0,"invalid":0,"nonfinite":0,"finite":1462,"entities":1,"missing_rate":0.0,"quantiles":[4.0,8.0,91.25,255.0,416.75,737.95,862.7800000000002],"iqr":325.5}
t7 bike / cnt	Window 10	{"window_id":10,"start_tick":"1346487000000000000","end_tick":"1351749900000000000","cells":1426,"missing":0,"invalid":0,"nonfinite":0,"finite":1426,"entities":1,"missing_rate":0.0,"quantiles":[5.0,8.0,76.25,246.0,449.75,784.75,900.75],"iqr":373.5}
t7 bike / cnt	Window 11	{"window_id":11,"start_tick":"1351749900000000000","end_tick":"1357012800000000001","cells":1456,"missing":0,"invalid":0,"nonfinite":0,"finite":1456,"entities":1,"missing_rate":0.0,"quantiles":[2.0,5.0,37.0,147.5,278.5,562.25,688.45],"iqr":241.5}
t7 bike / cnt	Comparison 0 → 1	{"reference_window":0,"comparison_window":1,"reference_support":1383,"comparison_support":1448,"reference_iqr":74.0,"reference_span":["1293858000000000000","1299120900000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_dist… [retained JSON contains full detail]
t7 bike / cnt	Comparison 1 → 2	{"reference_window":1,"comparison_window":2,"reference_support":1448,"comparison_support":1462,"reference_iqr":141.0,"reference_span":["1299120900000000000","1304383800000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_dis… [retained JSON contains full detail]
t7 bike / cnt	Comparison 2 → 3	{"reference_window":2,"comparison_window":3,"reference_support":1462,"comparison_support":1449,"reference_iqr":237.75,"reference_span":["1304383800000000000","1309646700000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_di… [retained JSON contains full detail]
t7 bike / cnt	Comparison 3 → 4	{"reference_window":3,"comparison_window":4,"reference_support":1449,"comparison_support":1458,"reference_iqr":220.0,"reference_span":["1309646700000000000","1314909600000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_dis… [retained JSON contains full detail]
t7 bike / cnt	Comparison 4 → 5	{"reference_window":4,"comparison_window":5,"reference_support":1458,"comparison_support":1456,"reference_iqr":210.0,"reference_span":["1314909600000000000","1320172500000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_dis… [retained JSON contains full detail]
t7 bike / cnt	Comparison 5 → 6	{"reference_window":5,"comparison_window":6,"reference_support":1456,"comparison_support":1455,"reference_iqr":167.25,"reference_span":["1320172500000000000","1325435400000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_di… [retained JSON contains full detail]
t7 bike / cnt	Comparison 6 → 7	{"reference_window":6,"comparison_window":7,"reference_support":1455,"comparison_support":1460,"reference_iqr":174.5,"reference_span":["1325435400000000000","1330698300000000000"],"reference_entities":1,"comparison_entities":1,"unit":"count","thresholds_are_investigation_defaults":true,"missingness_change":0.0,"missingness_review":false,"cdf_dis… [retained JSON contains full detail]

20 of 3759 metric rows shown. 3739 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 6 / 8

Availability and outcome maturity

Undeclared availability remains unknown. Historical agreement concerns supplied coverage; it does not establish what a pipeline consumed.

Evidence / source	Measurement or rule	Retained result
t6 bike	features	["cnt","casual","registered","temp","hum","weathersit"]
t6 bike	meaning	Undeclared availability remains unknown; no timestamp inheritance.

2 of 2 metric rows shown. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 7 / 8

Chronological evaluation

No retained evidence applies to this topic in the declared scope.

0 of 0 metric rows shown. Presentation limits do not change analytical coverage.

Methodology and coverage

Evidence / source	Measurement or rule	Retained result
method	schema	1.4
method	methodology	temporal_reliability_v1
method	execution status	completed
method	aggregate scope	Every eligible declared record in required stages; no silent sampling.
method	search scope [0] / selected measurements	["cnt","casual","registered","temp","hum","weathersit"]
method	search scope [0] / unsearched scope	[]
method	search scope [0] / work observation visits	5672766
method	search scope [0] / comparisons	504
method	search scope [0] / selection	Requested measurements first, then declaration order. Detail retention can miss behavior-only exceptions.
method	search scope [0] / defaults validated universal defect criteria	No
method	internal detail omissions / bike:entity histories	0
method	internal detail omissions / bike:series effects	26
method	timezone / package	tzdata
method	timezone / version	2026.5
method	timezone / zones / America/New York	d7f2206b3a45989fc9ad63d558922532fa7352280d5f87176bf1db79cb1d1fa9
method	phase seconds / bike:ingestion	0.1070317
method	phase seconds / bike:normalization	0.2589598
method	phase seconds / required diagnostics	0.2502283
method	phase seconds / availability history evaluation	1.1875e-05
method	phase seconds / behavior	1.349313

20 of 34 metric rows shown. 14 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Decoded-batch controls bound accepted batches and application retention. Native pages, dictionaries and conversion buffers can allocate before checks. Estimates and workload limits are provisional engineering controls. RSS monitoring can overshoot between samples.

Analytical scope left unsearched, retained-detail omissions, and presentation omissions are separate. No independent-row significance testing, model fitting, split generation, automatic repair or universal temporal score is performed.

Retained witnesses and selected detail

100 of 146 retained detail rows available here. Print shows up to 20 rows; JSON preserves every retained record. These presentation omissions do not change searched scope.

Evidence	Detail	Witness / selected population
t1	Witness	{"_row":7307,"measurement":"casual","time_state":"ambiguous_local_time","ordinal":7307}
t1	Witness	{"_row":7307,"measurement":"cnt","time_state":"ambiguous_local_time","ordinal":7307}
t1	Witness	{"_row":7307,"measurement":"hum","time_state":"ambiguous_local_time","ordinal":7307}
t1	Witness	{"_row":7307,"measurement":"registered","time_state":"ambiguous_local_time","ordinal":7307}
t1	Witness	{"_row":7307,"measurement":"temp","time_state":"ambiguous_local_time","ordinal":7307}
t4	retained histories	{"entity":[],"measurement":"casual","observations":17377,"cells":17377,"first_tick":"1293858000000000000","last_tick":"1357012800000000000","span_ticks":"63154800000000000"}
t4	retained histories	{"entity":[],"measurement":"cnt","observations":17377,"cells":17377,"first_tick":"1293858000000000000","last_tick":"1357012800000000000","span_ticks":"63154800000000000"}
t4	retained histories	{"entity":[],"measurement":"hum","observations":17377,"cells":17377,"first_tick":"1293858000000000000","last_tick":"1357012800000000000","span_ticks":"63154800000000000"}
t4	retained histories	{"entity":[],"measurement":"registered","observations":17377,"cells":17377,"first_tick":"1293858000000000000","last_tick":"1357012800000000000","span_ticks":"63154800000000000"}
t4	retained histories	{"entity":[],"measurement":"temp","observations":17377,"cells":17377,"first_tick":"1293858000000000000","last_tick":"1357012800000000000","span_ticks":"63154800000000000"}
t4	retained histories	{"entity":[],"measurement":"weathersit","observations":17377,"cells":17377,"first_tick":"1293858000000000000","last_tick":"1357012800000000000","span_ticks":"63154800000000000"}
t5	Witness	{"entity":"[]","measurement":"casual","first_absent_slot":"1248","end_absent_slot":"1254","absent_slots":"6"}
t5	Witness	{"entity":"[]","measurement":"cnt","first_absent_slot":"1248","end_absent_slot":"1254","absent_slots":"6"}
t5	Witness	{"entity":"[]","measurement":"hum","first_absent_slot":"1248","end_absent_slot":"1254","absent_slots":"6"}
t5	Witness	{"entity":"[]","measurement":"registered","first_absent_slot":"1248","end_absent_slot":"1254","absent_slots":"6"}
t5	Witness	{"entity":"[]","measurement":"temp","first_absent_slot":"1248","end_absent_slot":"1254","absent_slots":"6"}
t9	Witness	{"entity":"[]","first_tick":"1293858000000000000","last_tick":"1293858000000000000","observations":1,"state":"valid"}
t9	Witness	{"entity":"[]","first_tick":"1293861600000000000","last_tick":"1293861600000000000","observations":1,"state":"valid"}
t9	Witness	{"entity":"[]","first_tick":"1293865200000000000","last_tick":"1293865200000000000","observations":1,"state":"valid"}
t9	Witness	{"entity":"[]","first_tick":"1293868800000000000","last_tick":"1293868800000000000","observations":1,"state":"valid"}